Minor Project

Title:Cloud-Based AI System for Early Risk Detection of Chronic Disease Deterioration Using Daily Lifestyle and Behavioural Data

Presented by:
Hardik Lal: 500120516
Piyush Arora: 500121925
Shobhit Panchal: 500119775
Anshuman: 500119016

Mentored By: Dr Suresh Veesa

Content

Abstract

- WHO: Chronic noncommunicable diseases (diabetes, hypertension, heart disease, etc.) cause ~41 million deaths/year (≈71% of all deaths) , driven largely by modifiable lifestyle factors.
- Core idea: A Java/AWS cloud platform will analyse daily lifestyle/behavioural data (e.g. activity, diet, sleep) using ML models to predict early risk of chronic disease deterioration.
- Innovation: AI predictive analytics (e.g., as noted by Google Cloud) enable proactive interventions and early disease detection .
- Impact: Early warning alerts and personalised feedback can improve self-management, prevent complications, and reduce healthcare costs, aligning with WHO’s digital health goals.

Introduction

With the rapid growth of healthcare data and wearable technologies, predictive analytics and artificial intelligence are increasingly being used to detect health risks before they become severe. Machine learning models can analyze lifestyle patterns, physiological parameters, and clinical biomarkers to identify individuals who may be at risk of developing chronic diseases.

Traditional healthcare systems rely heavily on periodic medical examinations, which may delay the detection of underlying health issues. By integrating machine learning algorithms with healthcare datasets, it becomes possible to continuously monitor health indicators and predict potential risks early.

This project proposes a **Health Risk Prediction System** that analyzes patient lifestyle attributes and biomarker values to estimate the probability of future health risks. The system applies multiple machine learning models and selects the best-performing model based on evaluation metrics.

Algorithms Used

The project utilizes several machine learning algorithms for classification and prediction:

1. Random Forest
   - Ensemble learning technique using multiple decision trees
   - Reduces overfitting and improves accuracy
2. Gradient Boosting
   - Sequential learning algorithm
   - Each model corrects the errors of the previous model
3. Logistic Regression
   - Statistical classification model
   - Suitable for binary health risk prediction problems

The system also uses:

- Cross Validation (5-fold) for model reliability
- **ROC-AUC analysis** for model comparison
- **Confusion Matrix** for performance evaluation
- **Feature importance analysis** to understand influential health parameters

Problem Statement

Despite advances in digital health, most healthcare systems remain reactive, detecting chronic disease deterioration only after clinical symptoms worsen. There is a lack of cloud-based, AI-driven systems that continuously analyse daily lifestyle and behavioural data to provide early, actionable risk alerts for conditions such as hypertension, diabetes, coronary heart disease, and high cholesterol.

- **No unified cloud-based system for continuous early risk detection**
- **Lack of real-time AI-driven deterioration alerts**
- **Limited personalised risk monitoring using daily lifestyle patterns**
- **Fragmented data across health platforms**

Literature Review

Author	Year	Data Used	Methods / Models	Key Findings	Limitations
Chen et al.	2023	Simulated wearable datasets (activity, sleep, heart rate)	LSTM-based temporal prediction	Confirms lifestyle and wearable signals can predict early deterioration trends using simulated datasets	Wearable simulations may not capture real-world noise
MITRE Synthea Community	2024	Synthea synthetic health records	Population-level disease progression modeling	Widely adopted benchmark dataset for chronic disease simulation and early risk analytics	Not clinically validated for deployment
Kumar & Sharma	2024	Synthetic multi-condition cohorts (hypertension + diabetes)	Ensemble ML (Random Forest, XGBoost)	Demonstrates multi-disease deterioration modeling using synthetic cohorts	Limited transferability to real clinical settings
Google Health Research	2025	Simulated longitudinal patient monitoring data	Transformer-based sequence models	Achieves high performance in detecting deterioration patterns in simulated chronic disease trajectories	Black-box behavior; limited explainability
WHO Digital Health Sandbox	2025	Simulated patient monitoring datasets	Digital twin-based health simulation + ML risk scoring	Highlights the role of simulated patient data for testing early warning systems at scale	Not a substitute for clinical trials

SWOT ANALYSIS

Strengths

- Real-time risk detection
- Scalable cloud architecture
- Preventive healthcare approach

Opportunities

- Integration with wearable devices
- Deployment in hospitals and insurance platforms
- Government digital health initiatives

Weaknesses

- Dependence on data quality
- Limited accuracy without real-world continuous data

Threats

- Data privacy and compliance risks
- Ethical concerns in AI-based healthcare decisions

Objectives

To design and implement a **cloud-based AI system** that predicts early deterioration risk in chronic disease patients using **daily lifestyle and behavioural data**.

Sub-Objectives

- Build ML models for multi-disease risk prediction
- Ingest and preprocess lifestyle & health data in real time
- Deploy scalable AI services on cloud infrastructure
- Provide interpretable risk scores and alerts
- Ensure data security, privacy, and system reliability

Methodology

UPES
UNIVERSITY OF TOMORROW

A Software Model

The project follows an Agile-based iterative development model where model training, evaluation, and application development are performed incrementally.

Phase	Activity	Duration
Phase 1	Problem definition & literature review	1 week
Phase 2	Data preprocessing and model development	2 weeks
Phase 3	Model evaluation and optimization	3 weeks
Phase 4	Application development	3 weeks
Phase 5	Testing and documentation	2 week

Steps Involved

1. Data Collection
2. Data Preprocessing
3. Model Training
4. Model Evaluation
5. Best Model Selection
6. Application Development
7. Visualization

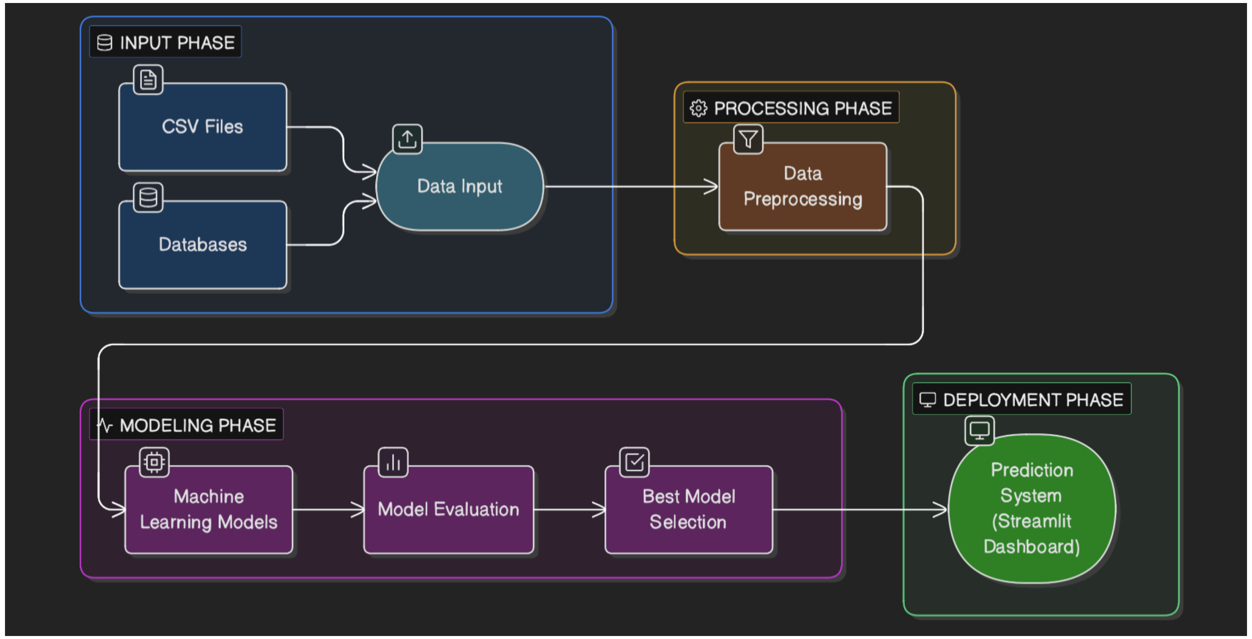
Working Model

Requirement Analysis

System requirements include:

Software Requirements

- Python
- Streamlit
- Scikit-learn
- Pandas
- Matplotlib



Hardware Requirements

- Standard computer system
- Minimum 8GB RAM recommended

UPES
UNIVERSITY OF TOMORROW

Conclusion

- The proposed cloud-AI system will enable early risk alerts for diabetes, hypertension, and heart disease based on daily lifestyle data.
- By leveraging scalable AWS services and ML, it offers a continuous monitoring solution, potentially improving patient self-management and enabling timely clinical interventions.
- Expected impact: Reduced progression of chronic diseases and lowered healthcare costs. As WHO highlights, tackling NCD risk factors can dramatically cut global mortality . This project aligns with WHO’s vision of equitable digital health and Google’s push for AI-powered personal health insights .
- Successful implementation could support public health goals by preventing complications and saving lives through proactive care.

Reference

- World Health Organisation (2026). Preventing noncommunicable diseases (NCDs): key facts and figures
- World Health Organisation. Global Strategy on Digital Health 2020–2027
- World Health Organisation. Harnessing Artificial Intelligence for Health (DG statement)
- Google Cloud. *Artificial intelligence (AI) in healthcare* (predictive analytics for risk detection)
- Google Health. *Advancing cutting-edge AI capabilities* (Large Sensor Model, PH-LLM for wearable data)
- Walonoski et al. (2018). Synthea: An approach, method, and software mechanism for generating synthetic patients... (MITRE)
- Al Mahmud et al. (2025). A Precision Health Service for Chronic Diseases using Wearable Devices... (JMIR mHealth & uHealth)

UPES

UNIVERSITY OF TOMORROW

Thank You